

Tesla Inc

Takeaways from Fremont Factory Tour, FSD Demos, and IR Meeting

We recently toured TSLA's Fremont factory and met with the IR team, Paresh Jain and Abhinav Davuluri, coming away with a deeper appreciation for the company's manufacturing capabilities, particularly the high degree of automation and operational intensity that defines the facility, while our discussion with IR provided incremental assurance around both the long-term vision and the near-term execution steps supporting it. Robotaxi scaling appears well positioned, with the limited Cybercab test fleet continuing to validate the technology, though the next major inflection in fleet deployment is tied to the roll out of FSD v15 later this year — TSLA also noted that further Model Y additions to the robotaxi fleet are being limited, reflecting management's conviction in the near-term scalability of Cybercab. On Optimus, production lines are being installed at Fremont (the area was tarped off at the time of our visit) and TSLA remains largely on schedule for the targeted 4 month transition following the May end-of-production for S/X, with initial humanoid deployments in 2H26 expected to focus on Optimus Academy for training and data collection, followed by internal factory use and external sales as early as 2H27. Gen 3 design is now finalized, locking in much of the supply chain, though esthetic elements are still being refined and will be revealed closer to SoP, and data collection remains the critical gating factor — as with FSD, TSLA is intent on building a proprietary in-house data repository. On FSD, the feature is increasingly becoming the primary purchase consideration for consumers, and while management sees long-term pricing power for current and future functionalities as exceeding the \$99/mo price point, the near-term focus is on expanding the subscriber base and driving usage, recognizing the importance of fueling both the sales and software development flywheels. Alongside growing FSD awareness and appreciation, TSLA's refreshed model lineup is also cited as a key catalyst for improving demand. Overall, we came away from the factory tour with greater conviction in the robotaxi fleet ramp through late-2026/early-2027, as well as an Optimus SoP that remains largely aligned with previously communicated timelines, both of which are central to the long-term thesis, while near-term commentary on demand drivers and FSD uptake remains constructive. Below are our detailed takeaways:

- **Facility details.** TSLA's Fremont facility serves as the company's primary test bed for innovation, supporting the launch of both new vehicles and humanoids across its expansive ~5 mn sq ft footprint. The plant manufactures premium Model Y and performance and premium Model 3 variants, while the recently decommissioned S/X lines (retired in early May) are being replaced by Optimus production lines. The factory showcases the breadth of TSLA's automation capabilities, featuring numerous custom-built robots and proprietary software designed to enhance manufacturing speed and precision. Production continues under several modernized tents, originally erected during the Model 3 production hell days, which remain integral to the facility's operations. During our visit, we observed gigapress machines, stamping operations, general assembly lines, and participated in FSD demonstrations, including with the Model Y L variant and the Cybertruck.
- **Giga casting and stamping.** TSLA's innovative giga casting process, first

Neutral

TSLA, TSLA US
Price (17 Aug 26):\$339.30

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introduced with the Model Y, consolidates 70 underbody components that would otherwise require welding, significantly streamlining production. This approach reduces the number of robots needed on the welding line — Model Y utilizes ~300 robots compared to ~1,000 for the Model 3. TSLA also attributes this advancement to its material science expertise, including the development of a custom aluminum alloy, a capability enhanced through its collaboration with SpaceX. On stamping, TSLA employs Schuler presses with six progressive, interchangeable die casts. TSLA stamps ~10 visible vehicle components in-house, while the remaining (hundreds) of stamped parts are sourced from suppliers.

- **Unboxed manufacturing.** TSLA emphasized that Cybercab production leverages an unboxed manufacturing process, where large subassemblies are built independently and in parallel, then joined together at the final stage. This approach allows open access from all angles, enabling simultaneous installation and streamlining the overall assembly process.
- **FSD Demos.** We participated in FSD demos with the Model Y L and Cybertruck, with the team running us through the various modes (ranging from Sloth to Mad Max) where the vehicle demonstrated notable driving accuracy, including navigating through a construction zone with no manual intervention. The Autopark function worked seamlessly as well.
- **Cybercab ramp up is closely tied to the FSD v15 roll-out.** Management expressed optimism about the trajectory and pace of the Cybercab production ramp, highlighting that technology validation and production (unboxed manufacturing) are advancing in parallel, with the fleet deployment timeline largely dictated by the upcoming FSD v15 launch later this year. TSLA indicated it is intentionally holding back on adding Model Y units to the robotaxi fleet, expressing confidence in its ability to scale the Cybercab fleet in the near-term. On FSD v15, TSLA views this release as a step-change in performance, comparable to the leap from v13 to v14, which saw a significant increase in parameter count, a larger context window, and a ~20% reduction in latency. The v15 upgrade encompasses seven core technologies, with ~40% of those currently being tested in the robotaxi fleet, where initial feedback has been encouraging. TSLA continues to focus on minimizing regression in core driving functions as it introduces new functionalities to the system, and FSD v15 is seen as the primary gateway to scaling unsupervised FSD. While the current AI/HW4 stack is capable of running v15 and supporting unsupervised FSD, TSLA's AI4.5 compute system is designed to future-proof against rising compute (~10% higher FLOPS) and memory (~2x higher) demands as robotaxi models scale and context windows expand.
- **Robotaxi unit economics remain compelling with Model 3/Y though Cybercab is designed to broaden TAM beyond traditional rideshare.** TSLA highlighted that current robotaxi economics are attractive, with Model Y and Model 3 total cost of ownership at ~\$0.60-0.70/mile at average personal vehicle utilization, and dropping ~\$0.50-0.60/mile at the 4-5x higher utilization rates typical for robotaxis, well below the ~\$2.5-3.0/mile price charged by incumbent rideshare operators. However, TSLA's vision for robotaxis extends beyond the traditional rideshare segment, which represents only a LSD% of the overall mobility market, and calls for a purpose-built robotaxi platform capable of driving TCO down to ~\$0.30/mile. Management also reiterated that the Cybercab is just the initial form factor, with additional vehicle types expected to follow as the platform evolves (citing the obovan demo from the 10/10 event as an example).
- **Optimus is on track for SoP in the coming months with commercial sales expected as early as 2H27.** While we did not have the opportunity to view the Optimus production lines currently being installed following the S/X decommissioning, TSLA indicated it remains largely on schedule with the ~4 month transition target (S/X production ended

in May 2026). No Optimus robots have been deployed at the Fremont factory yet, though management highlighted that stamping and body-in-white operations are likely to benefit most in the near term, given the repetitive and hazardous nature of those tasks. In contrast, the general assembly line still requires human dexterity and is expected to be a longer-dated application for humanoids. The initial rollout plan calls for Optimus robots to be deployed in the Optimus Academy in 2H26, where humanoids will learn by interacting with real-world environments, accelerating TSLA's data collection flywheel. Following this phase, humanoids will be deployed internally at TSLA factories for further data collection and to minimize third-party data complexities, with commercial sales to external customers anticipated as early as 2H27. TSLA is aligning humanoid data collection with FSD methodology and is committed to retaining data ownership internally, rather than relying on third-party datasets, to preserve long-term differentiation. Separately, TSLA noted that compute capacity has scaled ~2x y/y in 1H26, and design finalization (although aesthetics remain a work in progress) represents a key milestone for Optimus, signaling the company's commitment to scaling volumes longer-term (~1 mn units targeted at Fremont, with a longer-term goal of ~10 mn units at Giga Texas). The Gen 3 unveil will be timed closer to SoP to protect competitive advantage, while the scope of Gen 4 (in terms of capability, cost, and scalability) will be informed by Gen 3 field experience.

- **Demand rebound attributed to FSD and model lineup refreshes.** Excluding any potential EV demand uplift from higher gas prices tied to the Middle East conflict (which public franchise dealer management has indicated has not materially impacted BEV demand), TSLA attributes the recent acceleration in unit volumes (2Q26 unit sales +25% y/y and +34% q/q, largest q/q pickup since 2019) to progress on FSD and a refreshed model lineup. The introduction of new base variants, alongside the Model Y L and updated performance models, has broadened TSLA's offering, enabling the company to address a wider range of use cases and price points. On FSD, TSLA has made it the centerpiece of its product portfolio, leveraging improved functionalities to drive incremental customer interest, and management noted a growing trend of customers visiting showrooms specifically to learn about FSD. This sales momentum is also evident internationally, with markets such as Australia and South Korea, and early results from Europe showing a step-change in demand following FSD rollout.
- **Europe FSD approval update.** TSLA is pursuing a dual-track strategy for FSD approval in Europe, engaging directly with the EU (where the approval timeline has been repeatedly delayed and is now expected in October) while simultaneously working with individual countries such as the Netherlands, whose regulatory frameworks can be adopted by other EU member states. Early driving data from Europe shows promising safety metrics, including ~5x fewer collisions over ~65 mn kms of FSD driving, which management believes is gradually increasing regulatory momentum. Once approved, the activation timeline in European markets is expected to be relatively short, measured in weeks rather than months or quarters.
- **FSD pricing remains iterative in the near-to medium-term, albeit representing a significant long-term unlock.** TSLA explained that the transition from upfront FSD purchases to a subscription-only model is designed to maintain flexibility and capture the increasing value of ongoing functionality improvements. In the near term, management is prioritizing vehicle activations and FSD usage, especially as ~50% of TSLA owners historically have not tried FSD and also since a substantial cohort of customers who previously accessed older versions have not activated new subscriptions. TSLA continues to see strong stickiness once customers experience the technology, supporting the 1-month free trial for all new customers and the decision to hold pricing at ~\$99/mo.
- **Automotive gross margins.** TSLA noted it has made targeted pricing adjustments on certain Model Y variants and a broader set of Model 3 variants globally to help offset commodity cost headwinds, alongside changes to interest rate subvention that should

alleviate some gross margin pressure. Management also noted that 1Q26 and 2Q26 gross margins were modestly impacted by a shift toward subscription-based FSD monetization, as upfront FSD purchases ended in the US and Canada in February 2026 and will phase out elsewhere by August 2026. The ramp-up of cathode and anode production, which began in January 2026, is expected to gradually contribute to COGS efficiencies (typically 18 months for any plant to reach proper scale/utilization), though the magnitude may be difficult to isolate given several other moving parts. Broadly, TSLA emphasized that its operating philosophy is focused on expanding the top line, leveraging significant production runway (up to 3 mn production capacity targeted vs. JPMe 2026E deliveries of ~1.8 mn).

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Tesla Inc (TSLA, TSLA US) Price Chart



Date	Rating	Price (\$)	Price Target (\$)
03-Oct-23	UW	251.60	135
26-Jan-24	UW	182.63	130
03-Apr-24	UW	166.63	115
03-Oct-24	UW	249.02	130
24-Oct-24	UW	213.65	135
12-Mar-25	UW	230.58	120
23-Apr-25	UW	237.97	115
03-Oct-25	UW	436.00	150
29-Jan-26	UW	431.46	145
05-Jun-26	N	418.45	475
23-Jul-26	N	374.01	445

Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
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